

Formula Sheet — Test 2

Economics of Tourism | Universidade Europeia | Paulo Fagandini

Lectures 10–16: Producer Theory, Elasticity & Equilibrium | Lectures 18–26: Game Theory & Macroeconomics

Part I — Producer Theory, Elasticity & Equilibrium (Lectures 10–16)

1. Cost Concepts (L10)

$$\begin{aligned} TC &= FC + VC && \text{Total Cost} \\ AFC &= \frac{FC}{Q} && \text{Avg. Fixed Cost} \\ AVC &= \frac{VC}{Q} && \text{Avg. Variable Cost} \\ ATC &= \frac{TC}{Q} = AFC + AVC && \text{Avg. Total Cost} \\ MC &= \frac{\Delta TC}{\Delta Q} = \frac{\Delta VC}{\Delta Q} && \text{Marginal Cost} \end{aligned}$$

Key property: MC crosses AVC and ATC at their **minimum** points.

AFC always falls as Q rises.

2. Profit Maximisation (L11)

$$\begin{aligned} TR &= P \times Q && \text{Total Revenue} \\ \pi &= TR - TC && \text{Economic Profit} \\ \pi &= (P - ATC) \times Q^* && \text{Profit at } Q^* \\ MR &= P && (\text{perfect competition}) \end{aligned}$$

Golden Rule: produce where $P = MC$ (rising portion of MC).

$P > MC \rightarrow$ expand; $P < MC \rightarrow$ contract.

$P > ATC$ positive profit

$P = ATC$ zero economic profit (breakeven)

$P < ATC$ loss (may still operate)

3. Shutdown & Breakeven Rules (L12)

Shutdown price

$$= AVC_{\min}$$

• $P \geq AVC_{\min}$: **operate** (even at a loss)

• $P < AVC_{\min}$: **shut down** ($Q = 0$)

Breakeven price

$$= ATC_{\min}$$

• $P \geq ATC_{\min}$: non-negative profit

Intuition: shut down only if operating **worsens** the loss beyond FC . If $P > AVC$, revenue covers variable costs and contributes to FC .

4. Supply, Producer Surplus & Market Supply (L13–14)

The individual **supply curve** = MC above $AVC_{\min}$.

Producer Surplus (PS) — area above supply, below price:

$$PS = TR - VC = \frac{1}{2} \times Q^* \times (P^* - c)$$

(linear supply $P = c + dQ$; c = intercept)

$$\pi = PS - FC \quad \text{Profit from PS}$$

$$PS = \pi + FC \quad \text{PS from profit}$$

Market Supply (horizontal summation, n identical firms):

$$Q^{\text{market}} = n \times q_i(P)$$

Invert each firm's supply to get $q_i(P)$, then multiply by n .

5. Elasticity (L15)

Price Elasticity of Demand:

$$\varepsilon_d = \frac{\% \Delta Q_d}{\% \Delta P} \quad \varepsilon_d = \frac{1}{\text{slope}_P} \times \frac{P}{Q}$$

Price Elasticity of Supply:

$$\varepsilon_s = \frac{\% \Delta Q_s}{\% \Delta P} \quad \varepsilon_s = \frac{1}{\text{slope}_P} \times \frac{P}{Q}$$

Arc (midpoint) elasticity — works for both:

$$\varepsilon = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \div \frac{P_2 - P_1}{(P_1 + P_2)/2}$$

Supply shortcut (linear supply $P = c + dQ$):

$$\varepsilon_s = \frac{P}{P - c} \quad \begin{cases} c > 0 \Rightarrow \varepsilon_s > 1 \\ c = 0 \Rightarrow \varepsilon_s = 1 \\ c < 0 \Rightarrow \varepsilon_s < 1 \end{cases}$$

Demand: revenue test (price increase):

$|\varepsilon_d| > 1$ (elastic): $TR \downarrow$ $|\varepsilon_d| = 1$: TR unchanged $|\varepsilon_d| < 1$ (inelastic): $TR \uparrow$

Tax incidence: burden falls more on the **less elastic** side.

6. Market Equilibrium & Welfare (L16)

Equilibrium: set $Q_d = Q_s$ (or $D = S$) and solve.

Consumer Surplus (CS) — area above price, below demand:

$$CS = \frac{1}{2} \times Q^* \times (P_{\max} - P^*)$$

where $P_{\max}$ is the demand vertical intercept.

Total Welfare: $W = CS + PS$

Competitive equilibrium **maximises** W . Taxes, price ceilings, and price floors create **deadweight loss** (DWL).

Long-run equilibrium (perfect competition):

$$P = MC = ATC_{\min} \Rightarrow \pi = 0$$

• Profit > 0 : entry shifts supply right, P falls.

• Profit < 0 : exit shifts supply left, P rises.

Long-run supply is **horizontal** at $ATC_{\min}$.

Part II — Game Theory (Lectures 18–19)

7. Simultaneous Games & Nash Equilibrium (L18)

Three ingredients: players ∩ strategies ∩ payoffs.

Dominant strategy: best regardless of what the rival does.

Nash Equilibrium (NE): no player can improve by unilaterally deviating.

Finding NE — underline method:

1. For each column, underline the **row player's** highest payoff.
2. For each row, underline the **column player's** highest payoff.
3. A cell with **both** payoffs underlined = Nash Equilibrium.

Prisoner's Dilemma: both players have a dominant strategy; the resulting NE is **Pareto inferior** (both could be better off cooperating).

Escape: repeated games, reputation, tit-for-tat.

8. Sequential Games & Backward Induction (L19)

Backward induction: solve from the **last** node backwards.

1. Identify the last mover's optimal choice at each terminal node.
2. Fold back: the earlier mover anticipates this and chooses accordingly.
3. Repeat until the first node is reached.

Credible vs non-credible threats: a threat is **credible** only if carrying it out is **rational** when the time comes. Backward induction automatically filters out non-credible threats (bluffs).

Commitment: a sunk investment can make a threat credible by changing the payoff structure so that fighting becomes rational after entry.

First-mover advantage: exists when commitment to a large quantity/position gives the leader a structural edge (Stackelberg intuition).

Part III — Macroeconomics (Lectures 20–26)

9. Aggregation, GDP & Circular Flow (L20–22)

Expenditure approach:

$$Y = C + I + G + NX$$

C: consumption *I:* investment *G:* govt spending *NX:* net exports

Value-added approach: sum **value added** at each production stage (avoids double-counting).

$$\text{Value Added} = \text{Output price} - \text{Cost of inputs}$$

What is NOT in GDP: transfer payments (pensions, benefits); financial transactions; used goods; informal/non-market production.

Tourism in GDP: foreign tourist spending in Portugal = **export of services** (*NX*, credit).

Nominal vs Real GDP:

$$\text{Real GDP}_t = \sum P_{\text{base}} \times Q_t$$

GDP Deflator:

$$\text{GDP Deflator}_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} \times 100$$

Real GDP growth uses **real GDP** (base-year prices), not nominal.

10. Inflation & the CPI (L23)

CPI measures the cost of a **fixed basket** at current prices:

$$CPI_t = \frac{\text{Cost of basket at } P_t}{\text{Cost of basket at } P_{\text{base}}} \times 100$$

Inflation rate (year-on-year):

$$\pi_t = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \times 100\%$$

Common error: do not subtract CPIs directly — always divide by the starting CPI.

Real wage (purchasing power):

$$\text{Real wage}_t = \frac{\text{Nominal wage}_t}{CPI_t/100}$$

Three causes of inflation: demand-pull · cost-push · monetary.

Disinflation: inflation rate **falling** (prices still rising, but more slowly).

Deflation: price level **falling** ($CPI_t < CPI_{t-1}$).

CPI vs GDP Deflator: CPI = fixed basket (consumer goods); GDP Deflator = all domestic production (basket changes).

